

Erdős Problem #1122

EP-1122

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $f : \mathbb{N} \rightarrow \mathbb{R}$ be an additive function (i.e. $f(ab) = f(a) + f(b)$ whenever $(a, b) = 1$). Let $A = \{n \geq 1 : f(n+1) < f(n)\}$. If $|A \cap [1, X]| = o(X)$ then must $f(n) = c \log n$ for some $c \in \mathbb{R}$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>